Moriom Jerin Shithil

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EDUCATION

Khulna University of Engineering & Technology (KUET) Khulna, Bangladesh
Bachelor of Science in Electronics and Communication Engineering 2022–2026 | CGPA: 3.50/4.00

RAJUK Uttara Model College Dhaka, Bangladesh
Higher Secondary Certificate (HSC) 2018–2020 | GPA: 5.00

Bamna Asmatunnesa Pilot Secondary Girls School
Secondary School Certificate (SSC) 2018 | GPA: 5.00

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, SQL, Assembly, Verilog
Machine Learning & Artificial Intelligence: Deep Learning, Computer Vision, Convolutional Neural Networks (CNN), Swin Transformer, ConvNeXt, Transfer Learning, Explainable AI (XAI)
Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib
Engineering Tools: MATLAB, Proteus, Arduino IDE, Cisco Packet Tracer
Web & Research: HTML, CSS, PHP, Firebase, Scientific Writing, Literature Review, Experimental Design

RESEARCH EXPERIENCE

Research Intern, Elite Research Lab June 2026–Present

- Conduct research in Artificial Intelligence, Computer Vision, and Deep Learning.
- Develop and evaluate deep learning models using PyTorch for biomedical image analysis.
- Perform literature reviews, experimental design, data preprocessing, result analysis, and research documentation.

PUBLICATIONS & RESEARCH

Leakage-Aware and Imbalance-Sensitive Deep Learning Framework for 12-Class RBC Morphology Classification 2026
Undergraduate Thesis | PyTorch, Computer Vision, Explainable AI

- Developed a leakage-aware deep learning framework for 12-class red blood cell (RBC) morphology classification using smear-level data partitioning to prevent data leakage.
- Implemented ConvNeXt and Swin Transformer architectures and applied Grad-CAM for explainable AI; evaluated performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
- Achieved 89.55% Accuracy and 86.87% Macro-F1 under strict smear-level evaluation.

AI-Enabled Bilingual Wearable System for Real-Time Remote Health Monitoring 2025
IEEE BECITHCON – Conference Paper. **DOI:** 10.1109/BECITHCON69222.2025.11503921

- Designed a low-cost IoT-enabled wearable healthcare system integrating a bilingual Bangla-English Progressive Web Application with AI-assisted healthcare support using Arduino Uno, ESP32, Firebase, and Gemini API.

Hospital-Integrated Wearable System with AI & Physician Dual Consultation 2025
International Biotechnology Conference, BRAC University – Poster Presentation

- Developed a hospital-integrated platform combining AI analysis with physician consultation, implementing cloud-based patient monitoring and historical health record management using Firebase.

Machine Learning and Deep Learning Approaches for Alzheimer's Disease Detection: A Comparative Review 2026
Review Paper – In Preparation

- Conducting a comprehensive review of recent machine learning and deep learning techniques, neuroimaging modalities, public datasets, and explainable AI techniques for Alzheimer's disease diagnosis.

PROJECTS

Cisco Campus Network Design & Simulation Cisco Packet Tracer | 2025

- Designed and simulated a scalable campus network infrastructure using VLANs, DHCP, dynamic routing, ACLs, DNS, and Web servers to enable secure and efficient communication.

Wedding Management Database System Oracle SQL | 2024

- Designed a relational database system with modules for client registration, venue booking, catering, guest management, employee records, and event scheduling using CRUD operations.

Kindergarten Management System PHP, HTML, CSS, JavaScript, Relational Database | 2024

- Developed a dynamic web-based management system featuring user authentication, administrative dashboards, course management, and backend database connectivity.

E-Commerce Cart Management System C++ | Object-Oriented Programming | Data Structures | 2023

- Developed a shopping cart and inventory system using linked lists, customer profiles, product search, billing, and order processing.

Boolean Expression Design using Verilog Verilog HDL | 2023

- Designed and simulated combinational logic circuits and verified their functionality.

AWARDS

- Merit-Based Scholarship, Khulna University of Engineering & Technology (KUET)
- Board Scholarships: Junior School Certificate (JSC) and Secondary School Certificate (SSC)

CERTIFICATIONS

- IEEE BECITHCON 2025 – Paper Presentation & Certificate of Appreciation
- International Biotechnology Conference 2025 – Poster Presentation Certificate
- Computer Networking: Hands-on Experience (EDGE-DSTS, ICT Division)
- Mastering C & C++ Programming: From Fundamentals to Advanced (Udemy)
- Introduction to Database Management Systems (Udemy)

REFERENCES

Md. Minhajul Islam Arnab
Assistant Professor
Electronics and Communication Engineering, KUET
Email: arnab@ece.kuet.ac.bd

Md. Kishor Morol
CEO & Research Scientist
Elite Research Lab
Email: kishor@elitelab.ai